PROMPT ENGINEERING

Teaching Notes for AI Baithak

AI Baithak :learning, Exploring and Building with AI

How to use these notes

These notes are designed to support a 60–90 minute introductory classroom session. Explain the concepts in your own words, show the example prompts, and ask students to improve a weak prompt. The aim is practical understanding—not memorizing long definitions.

Learning outcomes

- Define a prompt and prompt engineering.
- Recognize common prompting techniques and know when to use them.
- Identify the main parts of a well-designed prompt.
- Write a basic prompt for SEO tasks.
- Write a basic prompt to request a website security review.
- Understand that AI-generated output and code must be checked.

1. What is a prompt?

A prompt is the input we give to an AI system to tell it what we want. It may be a question, an instruction, background information, code, an image, or a combination of these.

A prompt is the way a user communicates a task to an AI model. The clearer the task and relevant context, the easier it is for the model to produce a useful response.

Example: weak prompt and improved prompt

Weak prompt:

```
Explain artificial intelligence.
```

Improved prompt:

```
Act as an AI teacher.
Explain artificial intelligence to a BS AI student.
Use simple language and include:
1. A definition
2. Three real-world applications
3. One practical example
```

Teaching point: The improved prompt tells the AI the role, audience, task, style, and expected content. A longer prompt is not automatically better; relevant and clear instructions matter.

2. What is prompt engineering?

Prompt engineering is the process of designing, testing, and improving prompts so that an AI model produces useful, relevant, and consistent results.

It is more than asking questions. It means thinking carefully about the task, providing the right context, setting constraints, and specifying the output we need.

Why is it useful?

- It reduces ambiguity and unnecessary guessing.
- It helps produce answers in a required format.
- It makes AI outputs more relevant to a particular user or task.
- It supports tasks such as coding, summarization, classification, content creation, and data extraction.

Example: asking for code

```
Weak: Build a website.

Improved:
Act as a web developer.
Create a simple responsive portfolio website using HTML, CSS, and JavaScript.
Include an About section, Projects section, and Contact section.
Explain the files I need and how to run the website.
```

Teaching point: Prompt engineering improves the instructions; it does not guarantee that every AI response will be correct.

3. Common prompting techniques

A prompting technique is a particular way of structuring an instruction to help the AI handle a task. Different tasks may benefit from different techniques.

3.1 Zero-shot prompting

Give the AI a task without showing an example. Useful for straightforward tasks such as translation, summarization, and simple classification.

Example:

```
Classify this review as Positive, Negative, or Neutral:
"The delivery arrived two days late."
Return only the label.
```

3.2 One-shot prompting

Give one example so the AI can see the expected pattern or format.

Example:

```
Classify the review.

Example:
Input: "The staff were helpful."
Output: Positive

Now classify:
Input: "The product was damaged."
Output:
```

3.3 Few-shot prompting

Give two or more examples. This helps demonstrate a pattern, tone, or classification rule.

Example:

```
Classify each review.

"Excellent service." -> Positive
"Very poor quality." -> Negative
"It is acceptable." -> Neutral

Now classify:
"I am very happy with this purchase."
Return the label.
```

3.4 Role prompting

Tell the AI which role or perspective to adopt. The role helps establish the style and relevant expertise expected.

Example:

```
Act as a Python programming instructor.
Explain the difference between a list and a tuple.
Use a short code example and beginner-friendly language.
```

3.5 Contextual prompting

Provide background information that the AI needs before it performs the task. Context makes an answer more relevant.

Example:

```
Context: I am building a student portal using Python Flask and SQLite.
Task: Suggest three database tables needed for student registration.
Briefly explain each table.
```

3.6 Step-by-step / reasoning-oriented prompting

Ask for a clear explanation of the steps or calculations. This is useful for learning and checking problem-solving. It does not mean the model's explanation is guaranteed to reveal its internal reasoning.

Example:

```
A product costs Rs. 800 and is sold for Rs. 1,000.
Calculate the profit and profit percentage.
Show the calculation and final answer.
```

3.7 Structured-output prompting

Specify the exact output format, such as a table, list, or JSON. This is useful when another program will process the answer.

Example:

```
Extract the details from this message:
```

```
"My name is Ali. I need a haircut on Monday at 3 PM."

Return JSON with:
customer_name, service, date, time.
```

3.8 Decomposition prompting

Break a large task into smaller steps. This makes a complex task easier to plan, build, and review.

Example:

```
Help me plan a library management system.
First list the requirements only.
Do not write code until the requirements are reviewed.
```

3.9 Prompt chaining

Use a sequence of prompts where the result of one stage is used in the next stage. It is useful for multi-stage tasks.

Example:

```
Stage 1: Extract the skills from this resume.
Stage 2: Compare those skills with the job description.
Stage 3: Summarize the match and missing skills.
```

3.10 Multimodal prompting

Give the AI different kinds of input, such as text with an image, audio, or video, when the model supports it.

Example:

```
Look at the attached screenshot.
Identify the visible error message and suggest two possible causes.
Do not guess details that cannot be seen.
```

Note: These are practical introductory techniques. The best approach depends on the AI model, the task, and how the result will be used.

4. Anatomy of a professional prompt

A good prompt often contains six parts. Not every prompt needs all six, but they provide a useful checklist for more complex tasks.

Component	Question it answers	Example
Role	Who should the AI act as?	Act as a senior Python developer.
Context	What background does it need?	I am building a website for a university.
Task	What exactly should it do?	Design a student registration form.
Constraints	What rules or limits should it follow?	Use HTML, CSS, and JavaScript. Keep the design responsive.

Output format	How should the result be presented?	Provide the code in separate files and explain how to run it.
Quality criteria	What does a good result need to satisfy?	The form should validate required fields and show clear error messages.

Complete example combining the six parts

```
Role: Act as a frontend developer.
Context: I am creating a university website for students.
Task: Create a student registration form.
Constraints: Use HTML, CSS, and JavaScript. Make it responsive.
Output: Provide separate HTML, CSS, and JavaScript code.
Quality: Validate required fields and show understandable error messages.
```

Teaching point: Students should remember the purpose of each component, not memorize one fixed template.

5. Using prompt engineering for SEO

SEO stands for Search Engine Optimization. It means improving a website so that search engines can understand its pages and people can discover useful content through search results.

Basic SEO terms

Term	Basic meaning
Keyword	A word or phrase people type into a search engine, such as 'Python course for beginners'.
Search intent	The reason behind a search: learning, finding a website, comparing options, or buying something.
Title tag	The page title that can appear as the clickable heading in search results and browser tabs.
Meta description	A short summary that may appear below the page title in search results.
Headings	H1, H2, and H3 organize the content and make the page easier to understand.
Alt text	A text description of an image, useful for accessibility and when an image cannot be viewed.
Internal links	Links connecting one page of the same website to another.
Technical SEO	Technical improvements that help search engines crawl, understand, and access a website.

How can AI help with SEO?

- Suggest relevant keyword ideas.
- Identify likely search intent.

- Draft page titles and meta descriptions.
- Suggest a clear heading structure.
- Suggest related blog topics and internal links.
- Review page content for clarity and usefulness.

Example SEO prompt

```
Act as an SEO specialist.

Website topic: Python programming tutorials.
Audience: Beginners who want to learn Python.

Suggest:
1. Relevant keyword ideas
2. Search intent for each keyword
3. An SEO-friendly page title
4. A meta description
5. H1 and H2 headings
6. Five useful blog topics

Avoid keyword stuffing.
Do not invent facts or claim guaranteed rankings.
```

Important: AI can suggest SEO improvements, but it cannot guarantee a top Google ranking. Review suggestions and make content genuinely useful. Search engines value helpful, original content rather than content created merely to manipulate rankings.

6. Using prompt engineering for website cybersecurity

Cybersecurity means protecting websites, applications, systems, and data from unauthorized access, misuse, damage, and attacks. It is especially important when using AI coding tools because generated code can contain mistakes or security weaknesses.

Basic security terms

Risk	Basic meaning
SQL Injection	An attacker manipulates database queries through unsafe handling of input.
Cross-Site Scripting (XSS)	Malicious script content is injected into a page and runs in another user's browser.
CSRF	An attacker tricks a user's browser into sending an unwanted request to a site where the user is logged in.
Weak authentication	Login or identity checks are not strong enough to protect accounts.
Exposed API keys	Private credentials are accidentally made visible, for example in public frontend code.
Broken access control	A user can access information or actions they are not authorized to use.

Prompt injection	Malicious instructions in user input or external content try to manipulate an AI system.

Basic website security practices

- Keep private API keys on the backend; never place them in public frontend code.
- Store passwords using a suitable password-hashing algorithm; never store plain-text passwords.
- Use parameterized queries to help prevent SQL injection.
- Validate inputs and safely handle user-generated content.
- Check permissions on the server for every protected action.
- Use HTTPS and secure session and cookie settings.
- Keep frameworks and dependencies updated.
- Test the website before deployment and review AI-generated code.

Example cybersecurity review prompt

```
Act as a website security reviewer.

Review the code I provide.

Look for:
1. SQL injection
2. Cross-site scripting (XSS)
3. Weak authentication
4. Exposed API keys or secrets
5. Broken access control
6. Unsafe handling of user input

For each possible issue:
- Explain the risk in simple language.
- Identify the affected code if possible.
- Suggest a safer approach.
- Provide a corrected example where appropriate.

Do not claim the website is secure based only on this review.
```

Important: AI code review is not a security guarantee. Use proper testing, code review, and security tools. Only test websites and systems you own or have permission to assess.

7. Quick recap

- A prompt is the input or instruction given to an AI system.
- Prompt engineering is the process of designing and improving prompts.
- Techniques such as zero-shot, few-shot, role, contextual, and structured-output prompting solve different kinds of tasks.
- A professional prompt can include role, context, task, constraints, output format, and quality criteria.
- AI can assist with SEO research and content planning, but human review is still needed.
- AI can assist with security reviews, but generated code must be tested and checked independently.

8. Classroom activity

Ask students to improve this weak prompt:

```
Make me a website.
```

Ask them to include:

- Role: Who should the AI act as?
- Context: What is the website for?
- Task: What should it build?
- Constraints: Which technologies or rules should it follow?
- Output: What files or explanation are required?
- Quality: What should work correctly?

Optional follow-up: Ask students to write one SEO prompt and one cybersecurity review prompt for the website they described.

Suggested timing for a 60–90 minute class

Section	Suggested time
What is a prompt? + prompt engineering	10 minutes
Prompting techniques	20–25 minutes
Anatomy of a prompt	10 minutes
SEO prompt demonstration	10–15 minutes
Cybersecurity prompt demonstration	15–20 minutes
Student activity and recap	10 minutes